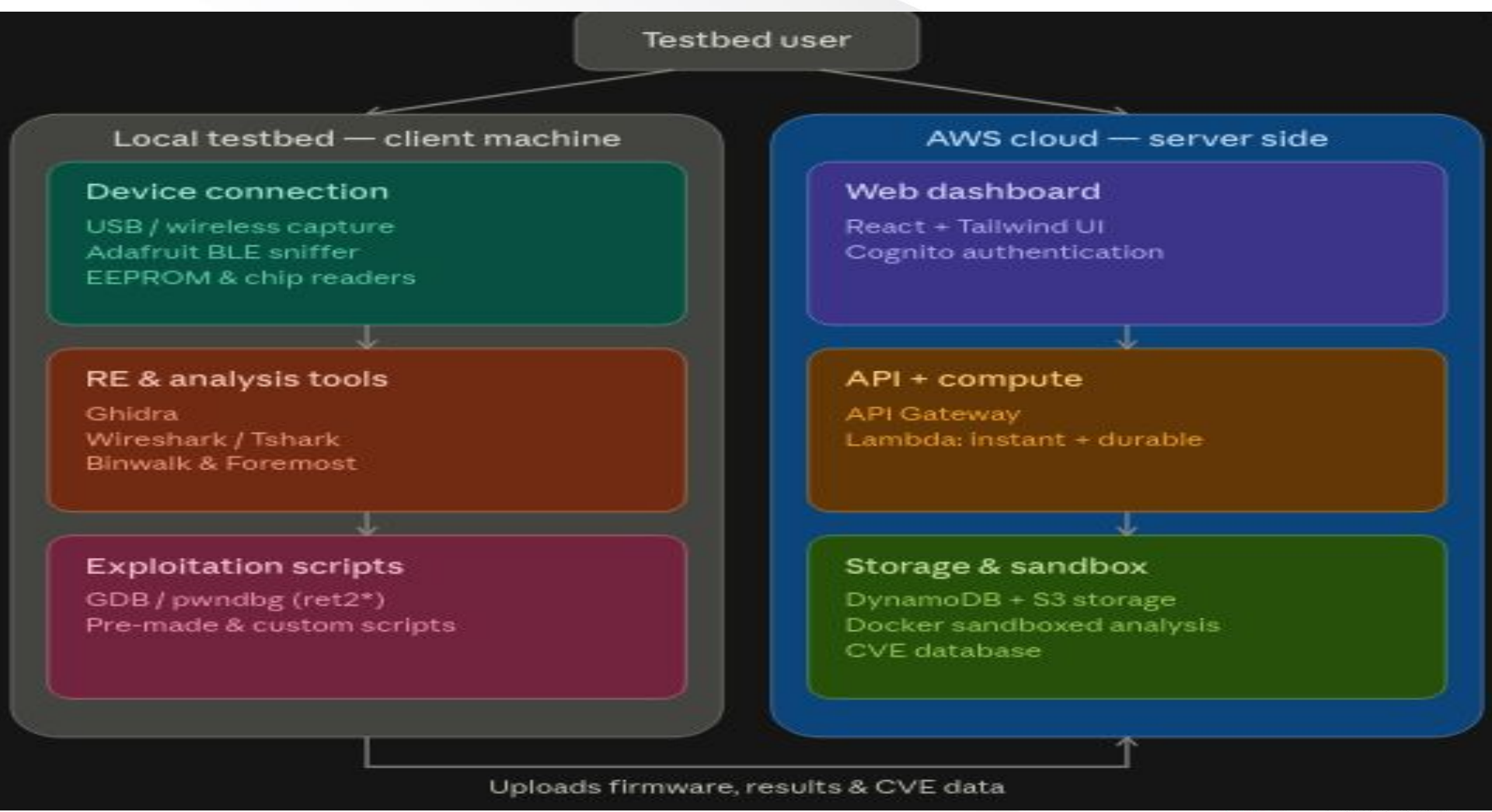
# Healthcare/IoT Cybersecurity Testbed: Sem. 2 Milestone 5

By: Justin Bower, Nathan Maloney, Ipule Pipi, and Jordan Chesley

Instructors/Advisors: Dr. Sudhakaran and Dr. Aydeger

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Testbed user
Local testbed — client machine
Device connection
USB / wireless capture
Adafruit BLE sniffer
EEPROM & chip readers
RE & analysis tools
Ghidra
Wireshark / Tshark
Binwalk & Foremost
Exploitation scripts
GDB / pwndbg (ret2*)
Pre-made & custom scripts
AWS cloud — server side
Web dashboard
React + Tailwind UI
Cognito authentication
API + compute
API Gateway
Lambda: instant + durable
Storage & sandbox
DynamoDB + S3 storage
Docker sandboxed analysis
CVE database
Uploads firmware, results & CVE data

# System Architecture Explained

Main groups:

1. Prospective Users: (Security Researchers, Manufacturers, Engineers, and Advisors)
2. User-facing systems: Web Dashboard (Server-side) and Local GUI (Client-machine)
3. Internal Systems:
   - AWS Cloud backend (includes DynamoDB, S3 Storage, Backend API, and VM Compute)
   - Standard commercial router flashed with OpenWRT and utilizing Tailscale or base Wireguard for tunnelling

Any internet access for testing will be provided by a dedicated ethernet line granted to us in the IoT lab.

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# Milestone 5 Progress Summary

| Task | % Complete | Justin | Nathan | Ipule | Jordan | To-Do |
|---|---|---|---|---|---|---|
| DS18B20 I2C Sensor/Buspirate Research & Integration | 50% | 25% | 0% | 0% | 25% | Continue research now that integration is complete |
| VR/RE & CVE database work | 100% | 60% | 0% | 0% | 40% | Look for our own CVES now that known ones are documented |
| Containerized Deployments & Platform Integration/Security Testing | 100% | 50% | 0% | 0% | | Perform additional platform stability/security testing now that div container/VM is deployable |
| Start-to-Finish Demo | 100% | 100% | 0% | 0% | 0% | None |

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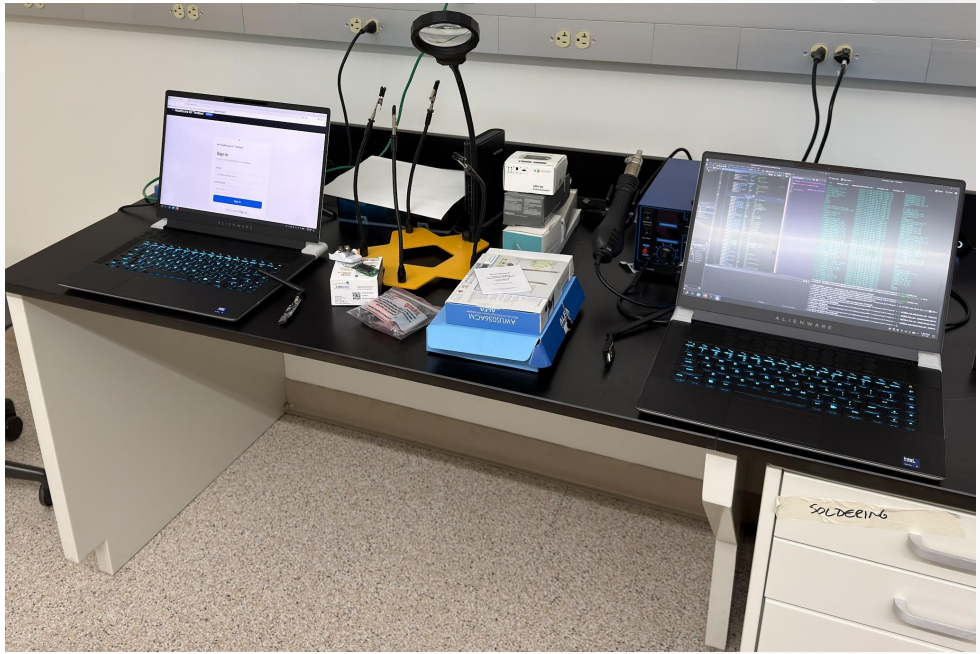
| Task | % Complete | Justin | Nathan | Ipule | Jordan | To-Do |
|---|---|---|---|---|---|---|
| Project Management, Communication, & Documents | 100% | 100% | 0% | 0% | 0% | Continue for next milestone |
| Cloud Platform Development, Collaborative Workspaces, UI/UX Redesign & Artifact Management | 100% | 0% | 0% | 100% | 0% | Fully implemented |
| Network Traffic Analysis Script Updates and Integration (Wireshark/Tshark) | 100% | 50% | 0% | 0% | 50% | Implement batch jobs/automation for NTA scripts |
| Firmware RE & Analysis Script Updates and Integration (Ghidra workflows, entropy analysis, Binwalk/Foremost+ segmentation) | 100% | 50% | 0% | 0% | 50% | Implement batch jobs/automation for Firmware/RE analysis & exploitation scripts |

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| Task | % Complete | Justin | Nathan | Ipule | Jordan | To-Do |
|---|---|---|---|---|---|---|
| Wifi/Bluetooth Exploitation Script Updates and Integration | 100% | 50% | 0% | 0% | 50% | Implement batch jobs/automation for Wifi/Bluetooth Exploitation scripts |
| Local GUI Development | 100% | 50% | 0% | 0% | 50% | None |
| Overall cloud improvements & python notebook editor | 100% | 0% | 100% | 0% | 0% | None |
| Cloud and Local GUI Integration | 50% | 0% | 40% | 0% | 0% | Implement firewall rules and deploy on a router |

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# Overall Testbed Progress Update




- This is the updated deployment of the physical testbed, featuring the virtual components, as follows:

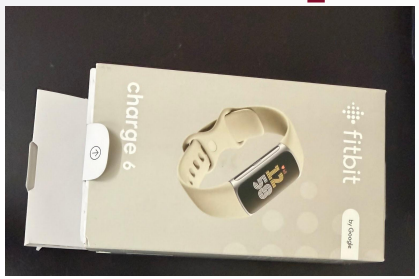
  - Server-side -> Web Dashboard (Left Laptop)
  - Local-Side -> Analysis Tooling (Right Laptop)

- We now be looking for our own CVES with the tool and implementing batch processing/automations
- We have researched/documented existing CVES for all of the devices

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# Heart Rate Monitor/Smartwatch RE Update

These were the devices RE’d this milestone. The documented chipsets and CVES are in a report on our public GitHub repo.







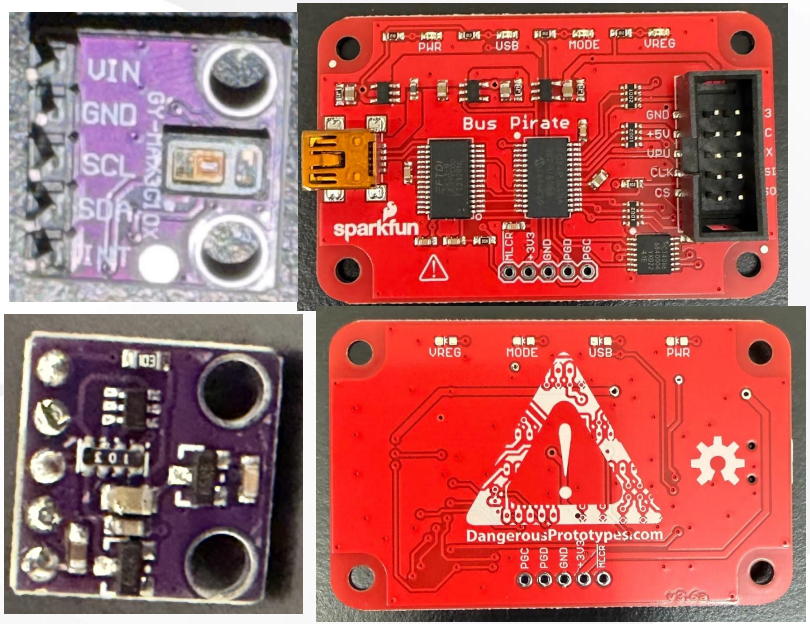




# I2C Sensor Research and Bus Pirate Review

To the right is the DS18B20 temperature sensor that have started researching.

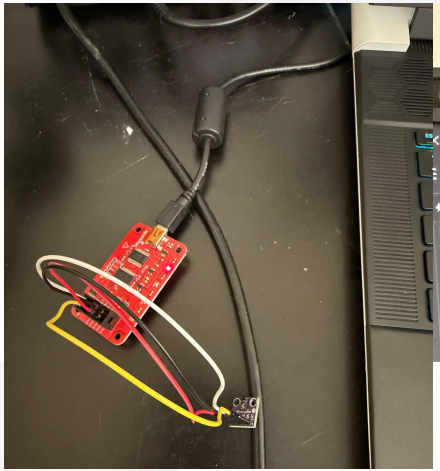
This sensor is usually found in oximeters,general vitals monitors, and other diagnostic medical devices.

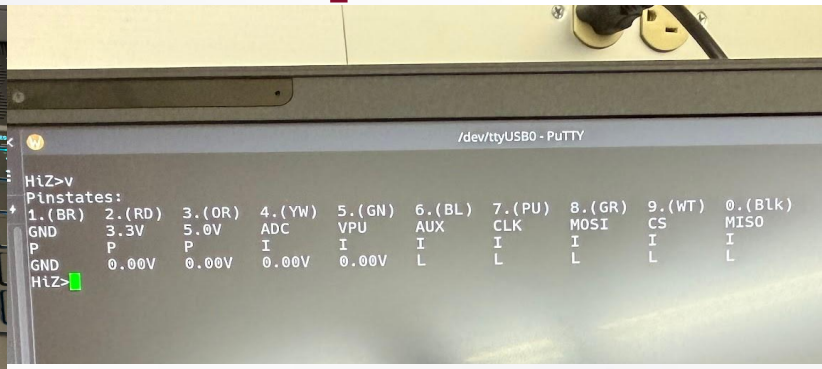


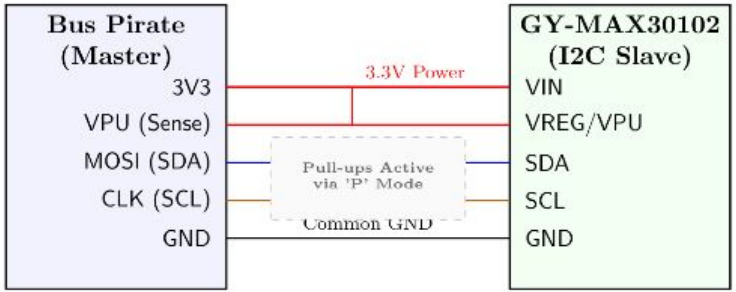

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# I2C Sensor Research and Bus Pirate Update

We used PuTTY to interface with the bus pirate, which was wired to the sensor, with a pinout diagrammed on the next slide.
The serial interface was /dev/ttyUSB0 and the baud rate was 115200 bps. More info is available on our public Github repo.








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# RE/VR Next Steps/Options

**Protocol Fuzzing and State Machine Subversion**

The device uses an internal 32-sample FIFO pointer array (0x04 to 0x07). Researchers can build scripts to fuzz unallocated or sensitive registers:

- **Out-of-Bounds Injection:**

  Rapidly writing unexpected values to the LED Pulse Amplitude Registers (0x0C, 0x0D) allows you to test whether the chip's power rails or internal state logic lock up, pull the shared data line down permanently (causing a Denial of Service), or trigger an unexpected hardware master clear.

- **FIFO Memory Exploits:**

  Overflowing the input buffers while manually altering read/write pointer boundaries (0x04 and 0x06) helps determine if the host microcontroller handles invalid pointer data gracefully or if it is susceptible to parsing errors.

**Man-in-the-Middle (MitM) Data Injection**

By intercepting and modifying packets on the fly, a researcher can spoof safe vitals metrics (ex. maintaining a simulated SpO2 of 98%) even if the sensor detects a drop.

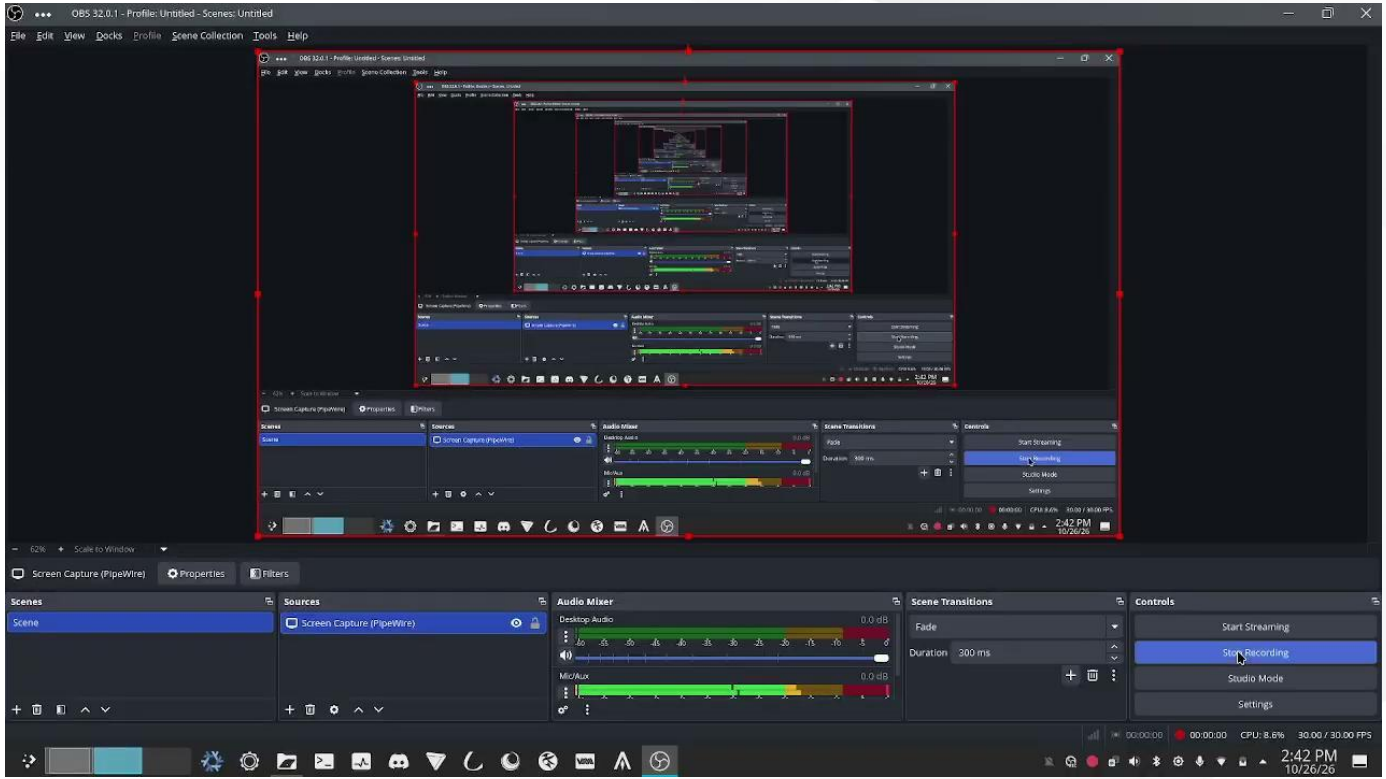
This allows you to evaluate how cleanly the upstream application validates data or handles protocol exceptions.

**Physical Side-Channel and Counterfeit Vulnerabilities**

- **Power Analysis:**
  Monitor current draw on the 3.3 V rail while the LEDs are pulsed. Distinct current signatures appear for different LED amplitude settings and can be used to fingerprint genuine versus counterfeit silicon.
- **Timing Side-Channels:**
  Measure the time between FIFO almost-full interrupt assertions and actual dataavailability. Variations can reveal internal state-machine behavior or manufacturing process differences.
- **Counterfeit Detection:**
  Compare the Part ID response, LED wavelength characteristics (via spectrometer if available), and I2C ACK timing against a known-good MAX30102 reference. Significant deviations indicate a cloned or relabeled part.

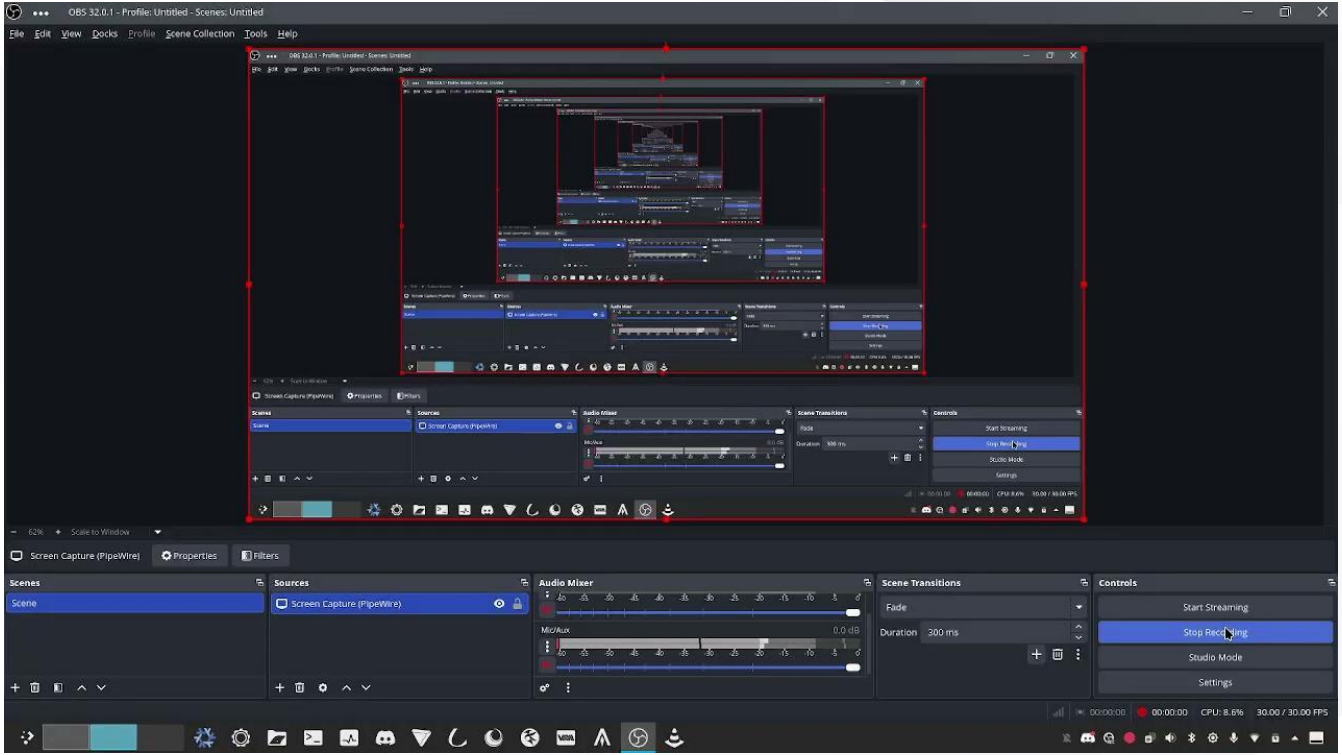
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# Local GUI Demo




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# VM Mode Demo




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← All runs

completed

Run another script on these outputs

# Firmware Static Inspector v1

Firmware Static Inspector v1 · economy S · started 10/8/2026, 12:58:49 PM by you

| WORK UNITS | JOBS | OUTPUTS | COST SO FAR |
|---|---|---|---|
| 1 / 1 | 1 / 1 | 1 | $0.0003 |
| 0 failed | 0 running | from 1 input file(s) | max $0.02 |

## Outputs

| | | | |
|---|---|---|---|
| firmware_report.json | log | 599 B | Download |

## Jobs

| JOB | STATUS | UNITS | ATTEMPTS | FINISHED | |
|---|---|---|---|---|---|
| #0 | succeeded | 1 | 1 | 10/8/2026, 12:59:29 PM | Log |

← Devices

# Alfa Network AWUS036ACM

Edit Device

Device · USB Wi-Fi Adapter · Workspace · **Healthcare IoT Testbed**

Your Access: Owner

Overview | Firmware 0 | Artifacts | CVEs 0

## Artifacts

Every file linked to this device, including firmware

Upload Binary

All | Firmware | Binary | PCAP | Logs | Kernel Modules | PDF Reports | Other

| NAME | TYPE | STATUS | SIZE | UPLOADED | ACTIONS |
|---|---|---|---|---|---|
| mt7662_rom_patch.bin | Binary | ready | 25.7 KiB | 10/24/2026, 12:40:37 PM | |
| mt7662u.bin | Binary | ready | 89.3 KiB | 10/24/2026, 12:40:37 PM | |
| mt7662u_rom_patch.bin | Binary | ready | 20.2 KiB | 10/24/2026, 12:40:37 PM | |
| mt7662.bin | Binary | ready | 80.0 KiB | 10/24/2026, 12:39:26 PM | |

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## Link CVEs

Choose recorded CVEs that affect **Alfa Network AWUS036ACM**.

375 | All severities | All devices | Clear filters | Refresh

| CVE ID | SEVERITY | CVSS | AFFECTED CHIPSETS | DEVICES | UPDATED | ACTIONS |
|---|---|---|---|---|---|---|
| CVE-2021-37566<br>WPS / IEEE 1905 mishandling resulting in an out-of-bounds write/... | High | 8.2 | MT7610UN MT7612UN | — | 6 Oct 2026 | Details |
| CVE-2021-37561<br>WPS / IEEE 1905 mishandling resulting in an out-of-bounds write/... | High | 8.2 | MT7610UN MT7612UN | — | 6 Oct 2026 | Details |
| CVE-2021-37562<br>WPS / IEEE 1905 mishandling resulting in an out-of-bounds write/... | High | 8.2 | MT7610UN MT7612UN | — | 6 Oct 2026 | Details |
| CVE-2021-37563<br>WPS / IEEE 1905 mishandling resulting in an out-of-bounds write/... | High | 8.2 | MT7610UN MT7612UN | — | 6 Oct 2026 | Details |
| CVE-2021-37584<br>WPS / IEEE 1905 mishandling resulting in an out-of-bounds write/... | High | 8.2 | MT7610UN MT7612UN | — | 6 Oct 2026 | Details |

Showing 5 of 16 CVEs

1 selected CVE-2021-37584 × Clear

Cancel | Link 1 CVE

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← Devices

# Alfa Network AWUS036ACM

Edit Device

Device · USB Wi-Fi Adapter · Workspace · **Healthcare IoT Testbed**

Your Access: Owner

Overview | Firmware 0 | Artifacts | CVEs 1

## Linked vulnerabilities

CVEs recorded in Healthcare IoT Testbed that affect this device

All vulnerabilities | Link CVEs

Linked 1 CVE to Alfa Network AWUS036ACM.

Search CVE, description, chipset, de | All severities | All devices | Refresh

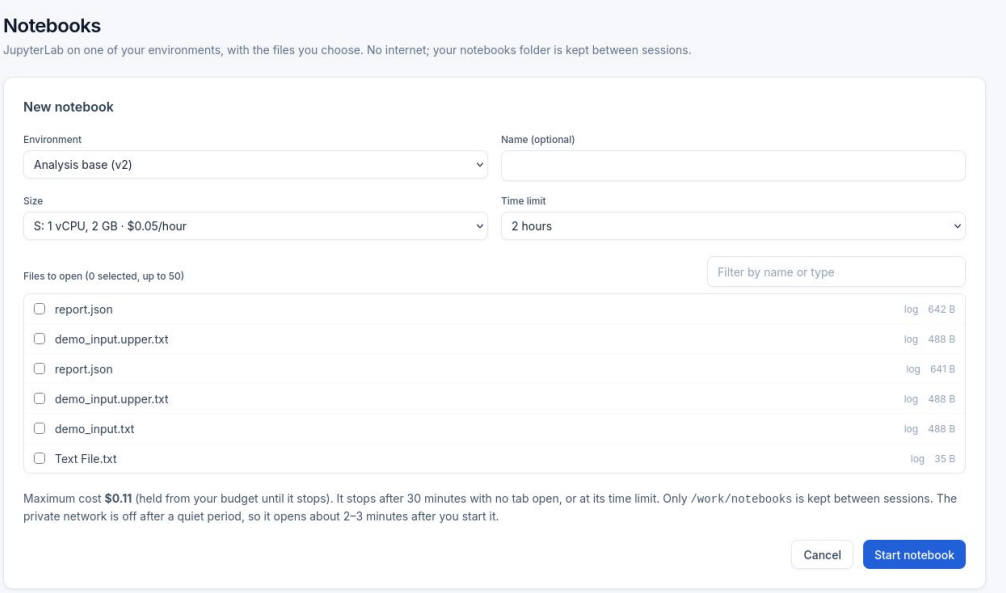
| CVE ID | SEVERITY | CVSS | AFFECTED CHIPSETS | DEVICES | UPDATED | ACTIONS |
|---|---|---|---|---|---|---|
| CVE-2021-37584<br>WPS / IEEE 1905 mishandling resulting in an out-of-bounds write/... | High | 8.2 | MT7610UN MT7612UN | Alfa Network AWUS036ACM | 26 Oct 2026 | ⋮ |

Showing 1 of 1 CVE

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# Jupyter Notebooks

- Users are now able to develop jupyter notebooks within the cloud environment in a secure session
- Sessions run on AWS Fargate with no internet access and no AWS credentials, and reach files only through the private manifest service
- Only the session's owner can open it, using a per-session token
- "Save as script" turns a notebook into an L1 script on the same environment




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# AWS Tech Debt and Future Planning

- A major rewrite of the backend was completed with the goal of improving the code base and allowing for a smooth handover of the project once the semester ends
- This was done by the transfer of all major configurations and variables to a config file that anybody can change and deploy
- All hidden AWS roles and permissions have been included in files within the repository to be manually populated into an AWS account




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# AWS/Web Dashboard Feature Eval. Review

Below is our evaluation for the server features:

- The testbed tightly integrates many disparate tools, automates firmware extraction, firmware analysis, entropy analysis, decryption, into one platform, is a novel approach to cybersecurity analysis.❌
- The testbed reduces the background knowledge needed for manufacturers or researchers to reverse-engineer Medical/IoT devices while unifying the reverse-engineering toolset.✔️100%
- The testbed is specifically tailored for IoT and medical hardware, which is often proprietary and notoriously difficult to analyze.✔️100%
- The testbed allows for direct firmware extraction from connected devices via standard interfaces and therefore streamlines the process of obtaining firmware/binaries for vulnerability assessment.✔️100%
- The testbed allows for built-in entropy analysis combined with pre-existing decryption techniques to uncover encrypted or obfuscated firmware sections more efficiently.✔️50%
- The testbed has support for both ready-made exploitation scripts and user-developed custom scripts provides flexibility to confirm known weaknesses or explore new attack vectors.✔️100%

We will list additional tests, especially for security, to ensure we are providing the maximum functionality.

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# Local GUI/User Testing Eval. Review

Below is was evaluation for the user tasks at the last milestone:

- Users can connect a variety of IoT and medical devices that use standard interfaces, wired or wireless, to the testbed, enabling full device access, interaction, and manipulation. ✔️ 100%
- Users can extract firmware directly from connected IoT and medical devices, obtaining complete firmware/binaries for detailed examination and vulnerability assessment.✔️ 100%
- Users can analyze firmware entropy to evaluate randomness and compression characteristics ✔️ 100%
- Users can utilize industry-standard open-source tools to perform reverse-engineering tasks and comprehensive vulnerability research. ❌
- Users can input their own CVE lists, and automatically retrieve lists of known CVEs, which are stored server-side in a database.
- Users can utilize pre-made scripts or upload custom exploitation scripts to rapidly test known , suspected, or new vulnerabilities, and potential attack vectors. ✔️ 100%
- Users can perform all connection, extraction, analysis, and exploitation tasks through an intuitive client-side graphical interface, thereby streamlining the process for both novice and experienced users. ✔️ 50%
- Users can create and upload documentation, or integrate existing documentation via GitHub, with all materials stored in server-side storage. ✔️100%

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# Local GUI/User Testing Eval. Update Part 1

- Users can connect a variety of IoT and medical devices that use standard interfaces, wired or wireless, to the testbed, enabling full device access, interaction, and manipulation. ✔️ 100%
- Users can extract firmware directly from connected IoT and medical devices, obtaining complete firmware/binaries for detailed examination and vulnerability assessment.✔️ 100%
- Users can analyze firmware entropy to evaluate randomness and compression characteristics ✔️ 100%
- Users can utilize industry-standard open-source tools to perform reverse-engineering tasks and comprehensive vulnerability research. ✔️ 100%
- Users can input their own CVE lists, and automatically retrieve lists of known CVEs, which are stored server-side in a database. ✔️ 100%
- Users can utilize pre-made scripts or upload custom exploitation scripts to rapidly test known , suspected, or new vulnerabilities, and potential attack vectors. ✔️ 100%
- Users can perform all connection, extraction, analysis, and exploitation tasks through an intuitive client-side graphical interface, thereby streamlining the process for both novice and experienced users. ✔️ 100%
- Users can create and upload documentation, or integrate existing documentation via GitHub, with all materials stored in server-side storage. ✔️100%

# Local GUI/User Testing Eval. Update Part 2

- Users can view and interact with GUI programs within the local GUI frame. ✔️ 100%
- Users can select files to be interfaced with by Local GUI tools/scripts. ✔️ 100%
- Users can add optional command-line arguments to dynamically run tools/scripts. ✔️ 100%
- Users can programmatically define their own tools/scripts to add to the Tools dropdown. ✔️ 100%
- Users can run multiple tools at once with different files and different arguments, can toggle between the different tools, and each tool has its own input/output. ✔️ 100%
- Users can run programs with ANSI escape code and see the processed output in the CLI view. ✔️ 100%
- Users can encounter errors in the Local GUI without the tool crashing ✔️ 100%
- Users can build a VM virtual disk for running the tool on any modern OS. ✔️ 100%
- Users can build a VM virtual disk quickly on any modern OS ❌
- Users can run the Local GUI locally or in a VM performantly ✔️ 100%
- Users can run the Local GUI,  in VM mode, on any modern OS, thus alleviating dependency issues.

# Milestone 6 Task Matrix

| Task | Justin | Nathan | Ipule | Jordan |
|---|---|---|---|---|
| Finalize CVE & RE Documentation | Finalize CVE entries & RE documentation for the final demo. | Assist with finalizing CVE entries & RE documentation. | Assist with finalizing CVE entries & RE documentation. | Assist with finalizing CVE entries & RE documentation. |
| Cyber Tooling Demos | Add demos of the cyber tooling into the final presentation. | None | None | Assist with preparing demos of the cyber tooling integrations. |
| Project Management & Comms | Keep team moving; maintain constant email/Discord communications with team & Professors. | None | None | None |
| AWS Backend Security & Integration | None | Finalize AWS permissions & backend security (least-privilege). Complete integration between local tools and cloud services | Assist Nathan | None |

| Task | Justin | Nathan | Ipule | Jordan |
|---|---|---|---|---|
| Dev Container Security Testing | Assist with testing. | Assist with testing. | Assist with testing. | Finalize & test containerized cloud analysis & isolation. Perform testing for final demo. |
| Local RE Pipelines Security Testing | Assist with testing. | Assist with testing. | Assist with testing. | Conduct testing of tool function security |
| End-to-End Web Workflow | None | Assist Ipule | Complete web workflow: device registration -> firmware upload -> analysis execution -> result retrieval. | None |
| Dashboard Integration & UX | None | Assist Ipule | Integrate RE/vulnerability outputs (entropy, discovered files, CVEs) into the dashboard. Organize devices, firmware, jobs, & outputs into a clear workflow. | None |
| Web App Edge-Case Testing | None | Assist Ipule | Perform edge-case testing (auth, ownership, upload retry, analysis initiation, results). Resolve integration issues & improve error handling. | None |

| Task | Justin | Nathan | Ipule | Jordan |
|---|---|---|---|---|
| Finalize Local GUI & Exploitation Scripts Batching Automation | None | Implement exploitation scripts batching & automation | Local RE pipelines by feeding results to the web backend. | None |
| Poster: Overall Design & Layout | Lead layout design & formatting; consolidate all sections into the final poster template. | Assist Justin with layout design, formatting, & consolidating sections. | Assist Justin with layout design, formatting, & consolidating sections. | Assist Justin with layout design, formatting, & consolidating sections. |
| Poster: Content Drafting & Visuals | Draft the RE/VR methodology, local GUI, & cyber tooling sections; provide workflow diagrams. | Draft cloud architecture & backend integration sections; provide diagrams/screenshots. | Draft web UI/UX & collaborative workspace sections; provide screenshots of the dashboard. | Draft the local pipeline & exploitation/analysis sections; provide screenshots of the Local GUI. |
| Poster: Overall Review & Polish | Proofread for consistency & formatting; ensure all required sections are present. | Verify technical accuracy of cloud/backend claims & diagrams. | Verify technical accuracy of web/dashboard claims & visuals. | Assist Justin with proofreading, formatting, & final polish. |

# Thanks for listening…
# Any questions?

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